

Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events.

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Abstract

Vehicles and pedestrians from major public events like sports, music, festivals, and faith meetings create massive numbers quickly, stressing transport systems. Traffic rises during these events causing bad congestion, poor signal timing, long delays, and hard flow maintenance. Even though Digital Twins monitor city traffic well, many tools focus on regular conditions and lack event-specific help. At the same time, traditional traffic improvement techniques fail to meet the massive, shifting, and multiple-goal characteristics found within transport networks during significant public gatherings. Thus, this paper proposes a Hybrid Quantum Traffic Optimization Framework Using Digital Twins for Major Public Events. A new Scheme for Traffic Simulation Setting and Planning Event Optimization via Digital Twin-based Traffic Simulation Environment and Hybrid Quantum-classical Optimization .This paper suggests a scheme joining Digital Twin-based traffic simulation setting with quantum-classical hybrid optimization to model, examine, and improve traffic states ahead of planned gatherings. Digital Twin models and produces the virtual setting of the transport network allowing road facilities, vehicle movement, traffic data, and

other relevant parameters, plus utilizes QUBO formulation to build a hybrid quantum-classical optimizer capable of modeling and improving traffic control steps regarding path diversion, traffic signal settings, and short-term road shutdowns. Transport officials may assess and choose the superior traffic control plan among various options employing the suggested Digital Twin-based traffic simulation setting within a simulator-based setting.

Keywords - *QUBO , QAOA , Ising Hamiltonian , Corridor analysis , Digital Twin , SUMO , TraCI , Non Homogeneous poisson process.*

1.INTRODUCTION

Large public gatherings involve a number of vehicles aggregating together leading to extensive congestion and extended travel distances. Even though the traffic is anticipated, its management is based on intuition, experience. Such techniques prove to be efficient but suboptimal therefore requiring smart flexible solutions heading for quantum computing providing a hope for addressing difficult combinatorial tasks more simply than traditional computers based on principles such as superposition hence identifying best answers to huge traffic control issues[9].

In classic systems, the basic unit of information used in computing and data transmission is a bit which can either be 0 or 1. On the contrary, quantum computing uses the qubit as its fundamental information. By definition, superposition dictates that qubits can be in multiple states at the same time i.e. they can be in both zero and one simultaneously. Superposition enables quantum computers to process an exponential number of possibilities at the same time on a large scale.

A Digital Twin is used to design a traffic environment, where traffic conditions can be compared to alternative optimized scenarios. In this way, a Digital Twin environment allows for both what-if analysis and assessment of performance relative to optimized alternatives. At the same time, quantum approaches formulate the traffic management problem as a combinatorial optimization problem. Since both route and traffic-control variables can make optimization a NP-hard task [13], the quadratic unconstrained binary optimization (QUBO) was chosen to encode decisions as binary variables and encode objective functions and penalties in a combined energy function. In this way, the binary variables indicate diversions for certain routes, extensions of certain junctions, and temporary restrictions of selected traffic routes.

The existing research has limited research considering the application of quantum optimization algorithms to event-based traffic management scenarios that require multiple coordinated decisions. The work by Yarkoni et al. [3] showed the ability of quantum computers to perform vehicle-route optimization for the Web Summit based on real-time traffic information, without considering coordinated optimization of multiple traffic-control decisions during the event. The use of quantum annealing for optimization has seen issues with scalability, connectivity, and other practical considerations for traffic management applications [6]. This motivates proposing a pre-event coordinated optimization framework that combines Digital Twin-based modeling and quantum-classical optimization of connected traffic-management decisions.

While Digital Twins have been studied extensively for quantum-optimized coordinated event-specific traffic management [2][7][8] simultaneous optimization of route diversions, signal extensions, and temporary

restrictions have received limited attention, with few demonstrations in practice [5][6]. As such, the coordinated optimization of multiple traffic-management decisions for event scenarios is an insufficiently explored area of research.

This work presents a pre-event quantum-classical traffic optimization framework for large public events, which encodes route diversions, signal extensions, and temporary restrictions as binary variables in a single QUBO and optimized using QAOA. A Digital Twin-based urban road network and pre-event synthetic demand is used to benchmark the resulting traffic-management plan against a classical approach, assessing coordinated pre-event planning capabilities, without making claims on quantum speedup or practical operationalization.

2. LITERATURE REVIEW

Yarkoni et al. [3] designed a bus-route management system for the 2019 Web Summit, issued in Lisbon: route selection was translated into a QUBO problem, which was solved on nine buses provided with D-Wave 2000Q processors. It proved the feasibility of the solution, which however was constrained by low levels of quantum resources and system scales. Kumarasamy et al. [2] experimented with an approach based on a Digital Twin and reinforcement-learning to signal control in an 11-intersection network in the MLK Smart Corridor; the approach lowered traffic delays and stops over traditional control, but it was limited to an 11-intersection network and was subject to information transfer delays.

Transportation. Mohammed et al. [6] surveyed quantum annealing applications for transportation such as traffic-signal control, traffic-flow optimization, and vehicle routing [11]. QUBO/Ising formulations were frequently used, but problems with qubit connectivity, hardware limitations, noise, and size persist. Faliagka et al. [8] developed a Traffic Monitoring and What-If analysis Digital Twin framework that takes data from sensors, vehicles, and applications; however, the approach was limited to a Digital Twin architecture, with technology integration, implementation costs, privacy, and infrastructure maturity as identified challenges.

Dribi et al. [5] performed a PRISMA systematic review of 165 papers on intelligent traffic management, found challenges such as scalability, data of different types, interoperability, latency,

privacy, and access to real-world validation. Kumarasamy et al. [7] extended the work to propose a Digital Twin-assisted decentralized multi-agent reinforcement-learning method for the optimization of traffic signal timing in an 11-intersection corridor, finding improvements in Eco-PI compared to actuated control but did not include quantum optimization.

Kim et al. [4] presented quantum annealing for large optimization problems reporting faster execution for some large-scale problems on hybrid quantum annealing but not in transportation-specific context. Prabakaran [1] proposed a FedQ-Twin by combining the concepts of federated learning, quantum-inspired optimization, and Digital Twin simulation of traffic management which needs further investigation of the real-world traffic data and physical infrastructure. In general, the previous work shows a progress of quantum optimization, Digital Twins, and intelligent traffic management[9], but there is still only a little research combined Digital Twin-based traffic simulation with quantum optimization for planning traffic-management strategies for a large public event. This research gap encourages the development of a hybrid quantum-classical framework for investigating the performance of event-specific traffic-management strategies in comparison with the actual traffic system.

3. MATERIALS AND METHOD

3.1 Dataset and Scenario

The assumed event based traffic-management scenario is an IPL match at the M. A. Chidambaram Stadium, with 4 key traffic corridors and a demand level of 9515 vehicles. The demand was artificially simulated based on an assumptions matrix derived from real-world based assumptions, taking into account road network configuration, traffic parameters, route choice options, limitations and signal extension levels. The event intensity, closure level, disruption level and demand multiplier were set at 0.8, 0.3, 0.2 and 1.2 respectively.

3.2 Hardware and Software

This work is implemented by using the Python 3.x, Qiskit 2.1.2, and Qiskit Aer 0.17.1 libraries. For data pre-processing and visualization, various libraries such as NumPy, Pandas, NetworkX, and Matplotlib have been used. The QAOA implementation was done using the statevector simulator of Qiskit Aer library.

3.3 Parameter settings

Parameter settings in the optimization experiment are summarized in Table 2. The classical brute-force approach used the same 18-variable formulation and sampled all $2^{18} = 262,144$ binary bitstrings. Random seed for the classical brute-force optimization was set to 42 and the same seed was used for all approaches as well. The QAOA circuit was evaluated with statevector simulator (without shot-based sampling).

Parameter	Setting
Number of binary decision variables	18
QAOA circuit depth (p)	2
γ values	$0, \pi/4, \pi/2$
β values	$0, \pi/4, \pi/2$
Number of parameter combinations	9
Quantum simulator	Qiskit Aer Statevector Simulator
Random seed	42

Table 1. Parameter Settings Used in the Optimization Experiment

4. Existing Method

An existing approach for event-based traffic optimization is the Quantum Shuttle framework proposed by Yarkoni et al. (2020). The framework addresses vehicle route assignment during large-scale events by formulating the route-selection problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem. Its objective is to assign routes to multiple vehicles while reducing overlap between their paths. The resulting QUBO problem is solved using quantum computing to obtain a suitable combination of vehicle routes.

4.1 Working Principle

Yarkoni et al. suggested the Quantum Shuttle technique for guiding vehicle routes for large events using a combination of transportation data, candidate-route generation, and QUBO-based optimization. The event demand determines routes, vehicle and traffic data determine candidate routes and are then filtered

based on travel times. All remaining routes are encoded as binary variables, optimized with QUBO, and returned for travel [3],[12] .

4.2 Limitations

The Quantum Shuttle framework demonstrated the use of quantum optimization for event-based vehicle routing, but the study also reported limitations related to quantum-hardware scalability, qubit connectivity, and noise. Frequent route updates could also cause route flickering, where vehicles receive different route recommendations before completing their current route. In addition, the formulation mainly considered overlap among the managed vehicle routes and did not directly include pedestrian or general event-attendee traffic [3].

4.3 Computational Considerations

The QUBO formulation involves pairwise interactions between route-selection variables, giving an approximate construction complexity of $O(n^2)$ for a dense representation. The study also reported response time associated with its cloud-based quantum service. These computational and scalability issues are relevant when applying the approach to larger event-based traffic problems [3].

5. PROPOSED SYSTEM

5.1 Quantum Optimization

The study developed a pre-event traffic planning framework for managing the anticipated vehicle movements related to a large public event. An IPL match at the M.A. Chidambaram Stadium, Chennai, was used as the experimental scenario. The framework is intended to help traffic authorities make traffic management decisions prior to the event by identifying appropriate combinations of route diversions, signal timing adjustments, and temporary traffic restrictions[10]. Therefore, the optimization is conducted as a planning exercise rather than as a real-time traffic control mechanism. A predefined event scenario, a modeled road network, and a set of operational constraints are utilized to evaluate various traffic management strategies. This approach allows different traffic management options to be compared before the event, helping authorities understand their possible effects on traffic flow. The results can then support the selection of practical strategies that reduce congestion and improve traffic movement during the event.

Quantum Traffic Optimizer - Overall Architecture

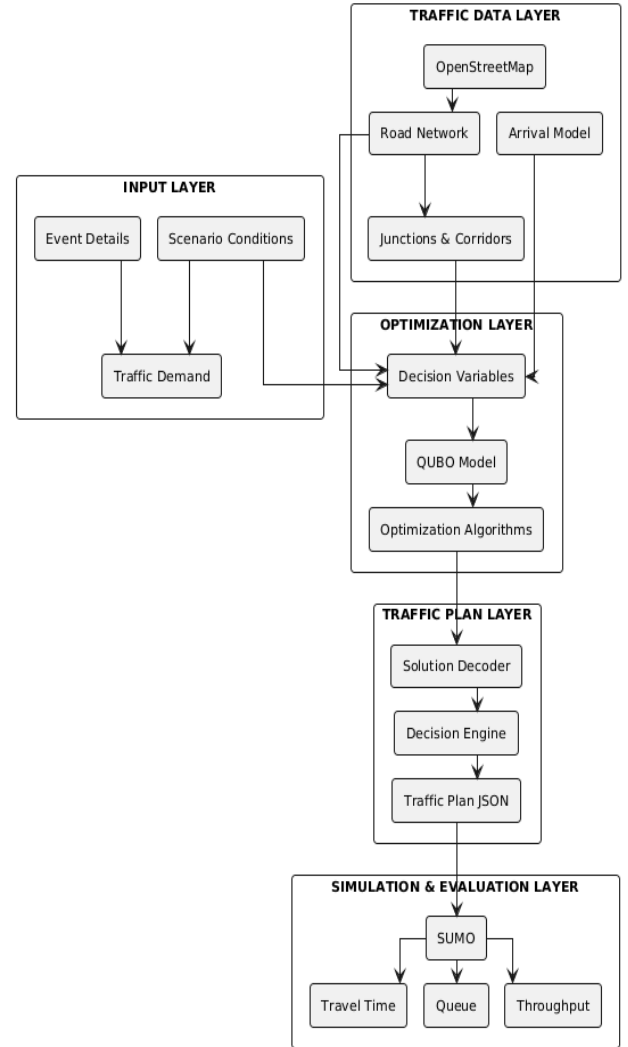


Figure 1: Overall System Architecture

In Figure 1 of Overall System Architecture the Experimental Scenario was established using total traffic-demand boundary of 9515 vehicles for the chosen event. Such criterion was selected as an experimental input on the scenario level to define a scale of a traffic condition imposed by the optimization framework. It must be emphasized that the selected value was not presented as a field-measured traffic count or a number of vehicles observed at the stadium. Instead, it establishes a general demand considered for the scenario, from which a particular part was explicitly defined for the selected network entry points, for the optimization experiment. This approach enables the study to evaluate the proposed optimization methodology under a consistent and controlled traffic-demand

condition while distinguishing the scenario input from empirical traffic data.

The experimental traffic demand was first determined on the level of the overall event scenario. Next, the demand was distributed to the selected entry points of the road network. In particular, 5 entry points were selected for explicit modeling in the event-area network, namely, J1-J5. Instead of using the number of vehicles that was found in the field measurements for these entry points, a certain traffic demand (in terms of number of vehicles) was set to each entry point J1-J5 in order to describe different levels of traffic loading at different approaches to the event area. In particular, the following demands were set: J1 – 1200, J2 – 1447, J3 – 600, J4 – 1100, J5 – 450. By setting different demands to different entry points J1-J5, the optimization model is no more using the same level of traffic loading for all approaches to the event area. Therefore, the demand that was explicitly set to each of the entry points J1-J5 is as follows:

$$D_{\text{modelled}} = D_{j1} + D_{j2} + D_{j3} + D_{j4} + D_{j5} \quad - \text{Eq(1)}$$

$$D_{\text{modelled}} = 1200 + 1447 + 600 + 1100 + 450 \quad - \text{Eq(2)}$$

$$D_{\text{modelled}} = 4797 \text{ vehicles} \quad - \text{Eq(3)}$$

The total scenario-level demand of 9515 was kept intact while 4797 was particularly allocated to the five modeled entry points. This distinction was made for the experimental demand not to be confused with either a traffic count that had been measured or observational data on some roads. The demand for the other scenarios was considered in larger experimental representation instead of being artificially distributed among the five entry points.

As the vehicles that are linked to the large public event would arrive with some timing, rather than all at once, the model was applied with a component accounting for this temporal arrival. A part of the vehicles linked with the event that comes during every interval was represented by a previously defined scenario-based profile. Let p_t represent the share of vehicles arriving during the t -th interval. The summation of the shares for the entire planning period is assumed to be limited by :

$$\sum_t p_t = 1 \quad - \text{Eq(4)}$$

The demand assigned to each time interval is consequently represented as

$$D_t = D_{\text{total}} p_t \quad - \text{Eq(5)}$$

where D_t represents the demand during time period t , D_{total} signifies the demand utilized in the temporal model and p_t expresses the ratio of that demand expressed in the corresponding period. The interpretation appears to be more similar to the behavioral model of such phenomena. The temporal profile has not been considered as a reliable source of empirical data because there are no evidence collected in the field about arrivals in the current experimental regime. The routing surrounding the event area was obtained from OpenStreetMap using OSMnx library in Python. In the practical experiment, the radius of the network was established to be approximately 1200 m away from M. A. Chidambaram Stadium. The acquired routing was illustrated as a graph with nodes being the intersections and edges being the representation of road segments linking the points. The graph in question consisted of about 750 points of intersection and 1985 road segments. By utilizing this graph representation, necessary information on the connections, routes and traffic flows in the event region was obtained.

In this optimization scenario, four key roads were selected as they were associated with the flow of traffic in the area of the event. The roads in question are Anna Salai, Triplicane High Road, Wallajah Road and Kamarajar Salai with number of intersections being 22, 70, 25 and 14 respectively. The roads in question were looked at as potential locations for route traffic interventions and the number of intersections was looked at as potential locations for modifying traffic signal timing. Therefore, the selection of the roads was based on the relevant configuration of the road network that were looked at in the event under quantification.

Traffic capacity was considered as a variable in the modeling. For the geographical and connectivity perspective of the network, OpenStreetMap was used, while the capacity figures utilized in the modelling process were treated merely as modelling variables rather than actual data from the field. The relationship between demand and capacity was modelled through the demand to capacity ratio

$$R = D / C. \quad - \text{Eq(6)}$$

Where D is the traffic demand and C is the associated capacity. If $R > 1$ then this means that the demand is higher than the capacity and if $R < 1$ then this implies that the capacity is more than the demand. The network average capacity and the corridor operational capacity were considered to be two different concepts because otherwise, it could have been misleading to consider the network average as the corridor capacity.

The traffic-management problem was formulated as a binary optimization problem in which every possible intervention was represented by a binary decision variable [14]. For each intervention i , $x_i \in \{0, 1\}$

$x_i = 1$ denotes selection of intervention i while $x_i = 0$ denotes non-selection of intervention i . In total, 18 decision variables were used. The first four decision variables, x_1 to x_4 , corresponded to route diversion decisions with respect to the four chosen corridors. The next ten decision variables, x_5 to x_{14} , corresponded to signal extension decisions for intersections J1 to J10 respectively [11][15]. The last four decision variables, x_{15} to x_{18} , corresponded to temporary restriction decisions with respect to the four chosen corridors [17]. Thus the decision variable vector could be written as

$$x = [x_1, x_2, \dots, x_{18}]^T \quad \text{-Eq(7)}$$

Operational constraints were introduced to limit the number of interventions that could be selected simultaneously. A maximum of two route diversions was permitted, expressed as

$$x_1 + x_2 + x_3 + x_4 \leq 2 \quad \text{-Eq(8)}$$

Similarly, the number of signal extensions was restricted to a maximum of five:

$$x_5 + x_6 + x_7 + x_8 + x_9 + x_{10} + x_{11} + x_{12} + x_{13} + x_{14} \leq 5$$

-Eq(9)

The number of temporary restrictions was limited to one:

$$x_{15} + x_{16} + x_{17} + x_{18} \leq 1 \quad \text{-Eq(10)}$$

The above constraints have been applied to ensure that the final traffic management strategy remains operationally feasible and does not select too many interventions at once for the optimization procedure.

The traffic management formulation took into consideration different dimensions of the resulting traffic condition such as travel time, length of queue, capacity, congestion-based variables, safety variables, and cost of implementation. The initial objective for traffic performance conceptually was represented as

$$F(x) = w_t T(x) + w_q Q(x) - w_r R(x) + P(x) \quad \text{-Eq(11)}$$

Quantum Traffic Optimizer - Optimization Architecture

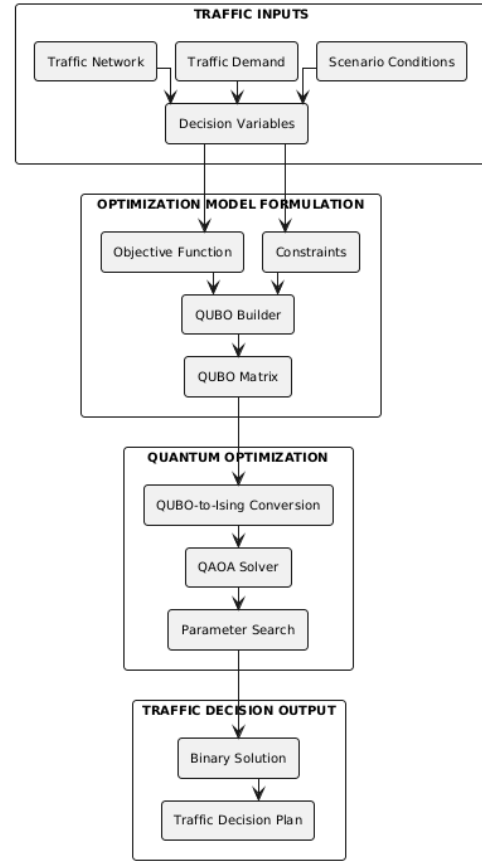


Figure 2: Quantum Optimization Architecture

In this equation, $T(x)$ represents the travel time, $Q(x)$ represents the costs resulting from queuing, $R(x)$ is the benefit from selecting the intervention, while w_t , w_q , and w_r are the weighting factors for each, respectively. The $P(x)$ represents the penalty for constraint violations.

The initial performance objective for traffic and the QUBO energy were two separate entities in this study's methodology. The traffic performance formula is based on the actual traffic management problem in operation, while the QUBO was the mathematical optimization model fed into the optimization phase. The QUBO that was formulated included the individual decision contribution, pairwise decision interactions, cost of implementation, scenario rewards, and penalties.

Converting the constrained binary optimization problem into the unconstrained quadratic problem was achieved using the approach that involved including the operational constraints within the objective

function using penalties. The QUBO formulated took the form

$$E(x) = \sum_i a_i x_i + \sum_{i < j} b_{ij} x_i x_j + P(x) \quad \text{-Eq(12)}$$

where a_i is the linear effect of the decision variable x_i , b_{ij} is the interaction term of the decision variables x_i and x_j , and $P(x)$ is the penalty effect for violating the constraints.

In the case of the applied traffic optimization problem, the QUBO model involved 18 binary variables that could be defined using a matrix of size 18 x 18. The formulation included 18 linear terms and 153 quadratic interaction terms with the use of penalty terms for three operational constraints. The QUBO matrix could be written as

$$E(x) = x^T Q x \quad \text{-Eq(13)}$$

where x is the binary decision vector and Q is the QUBO coefficient matrix. Diagonal entries in Q denote the contribution of the decision variables individually, whereas off-diagonal entries signify interactions between two decision variables. Quadratic interactions permit the inclusion of the information in the model that the effect of one traffic management strategy depends upon other traffic management strategies being included as well.

For the purpose of using the quantum optimization method, all the binary variables were converted into spin variables with the transformation

$$x_i = (1 - s_i) / 2 \quad \text{-Eq(14)[20]}$$

with

$$s_i \in \{-1, +1\} \quad \text{-Eq(15)}$$

The resulting formulation was expressed using an Ising Hamiltonian of the form

$$H_c = \sum_i h_i Z_i + \sum_{i < j} J_{ij} Z_i Z_j \quad \text{-Eq(16)}$$

Here, h_i is the weight associated with the individual variable, J_{ij} is the weight of interaction between the two variables, and Z_i is the Pauli-Z operator for the respective qubit. This conversion changes the binary optimization problem to the Hamiltonian format used in the QAOA algorithm [19]. The conversion of the optimization problem to the Hamiltonian format is a purely mathematical procedure.

Quantum optimization step was designed using Quantum Approximate Optimization Algorithm (QAOA) [16]. As there are 18 binary decision variables in the problem, 18 qubits were utilized in designing the quantum algorithm. An equal superposition state of the computational basis states was considered as the initial quantum state. An initial quantum state for a single qubit can be expressed as

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2} \quad \text{-Eq(17)}$$

The initial state for the complete 18-qubit system is therefore

$$|\psi_0\rangle = |+\rangle^{18} \quad \text{-Eq(18)}$$

The 18 binary variables generate a total search space of

$$2^{18} = 262,144 \quad \text{-Eq(19)}$$

possible binary configurations. The QAOA approach creates a process whereby these possible states can be assigned certain amplitudes that will allow the optimization process to lead to states with low cost Hamiltonian values.

Although exhaustive classical enumeration remains possible for the 262,144 configurations in our present 18-variable configuration, QAOA provides a quantum-based optimization that may be investigated on problems extending beyond this set. Its role in the present study is therefore to assess the applicability of quantum optimization to the formulated traffic-management problem, while the current size of the problem allows for direct comparison to classical enumeration and optimization.

The Ising Hamiltonian obtained from the QUBO was used as the cost Hamiltonian of the QAOA formulation. The corresponding cost unitary operator was defined as

$$U_c(\gamma) = e^{-i\gamma H_c} \quad \text{-Eq(20)}$$

where γ represents the variational parameter associated with the cost layer. A mixing Hamiltonian was defined using the Pauli-X operators as

$$H_M = \sum_i X_i \quad \text{-Eq(21)}$$

The corresponding mixing operator was defined as

$$U_M(\beta) = e^{-i\beta H_M} \quad \text{-Eq(22)}$$

where β represents the variational parameter associated with the mixing layer. For a single QAOA layer, the resulting state can therefore be expressed as [21]

$$|\psi(\gamma, \beta)\rangle = U_M(\beta)U_C(\gamma)|\psi_0\rangle \quad \text{-Eq(23)}$$

For p QAOA layers, the state is generalized as

$$|\psi(\vec{\gamma}, \vec{\beta})\rangle = U_M(\beta_p)U_C(\gamma_p) \cdots U_M(\beta_2)U_C(\gamma_2)U_M(\beta_1)U_C(\gamma_1)|\psi_0\rangle \quad \text{-Eq(24)}$$

The variables γ and β set the transformation performed by the cost and mixing layers. The fitness of a particular setting of the parameters can be assessed based on the expectation of the cost Hamiltonian:

$$\langle H_C \rangle = \langle \psi(\vec{\gamma}, \vec{\beta}) | H_C | \psi(\vec{\gamma}, \vec{\beta}) \rangle \quad \text{-Eq(25)}$$

This process tries to find settings where the expected cost is minimized and the likelihood of achieving low-energy binary solutions is maximized.

In the current approach, the QAOA solution was simulated classically via a grid search on the variational parameters. Therefore, the results obtained for the QAOA approach are simulation results for the QAOA formulation and not those of executing on physical quantum hardware. The current implementation enabled the formulated QAOA to be run with the stated traffic setup.

After the optimal state had been obtained through the optimization process, it was then decoded to determine the appropriate traffic management decisions. If a decision variable $x_i = 1$ for a route-diversion variable, it meant that the respective route had been chosen to be diverted. If a decision variable $x_i = 1$ for a signal-extension variable, it meant that the respective intersection had been chosen to be modified, and if a decision variable $x_i = 1$ for a temporary-restriction variable, it meant that the respective route had been chosen to be temporarily restricted.

The traffic model evaluated the resulting configuration using travel time, queue length, throughput, congestion-related measures, and implementation penalties [18]. Throughput represented the number of vehicles processed by the explicitly modeled network and was therefore distinct from the initial scenario demand of 9,515 vehicles. The implementation penalties were assigned as weights of 8, 3, and 12 for route diversion, signal extension, and temporary

restriction, respectively, with the total penalty determined by the selected decision variables.

The QUBO energy was not considered to be a direct measure of the performance of the traffic. The QUBO energy is the mathematical energy that is optimized in the process. On the other hand, traffic time, traffic queues, and all the other parameters are the result of traffic model calculations, after the chosen traffic management setting is analyzed using the traffic model. Thus, the QUBO objective and traffic performance parameters remained separate measures.

The QAOA-based formulation was benchmarked against two additional formulations of the optimization problem using the same traffic situation and decision problem formulation. The first alternative was based on the application of the greedy technique wherein interventions were performed on an individual basis based on expected gains from each intervention without the joint optimization of all interactions between decisions. The second approach was classical optimization where the same decision variables, objective formulation, and constraints were applied with the use of classical optimization procedures. The QAOA-based technique used the same formulation with QAOA-based parameter search procedure.

For the current experimental case, the greedy algorithm saved about 2.8 minutes of travel time and reduced the queue by 6%, taking about 0.016 seconds of computing time. For the classical optimization method, the travel time was saved by about 3.3 minutes, reducing the queue by 8%, with the computing time being about 2.55 seconds. For the QAOA-based method, the travel time savings was about 3.34 minutes, while the queue was reduced by 8.2%, taking about 18.03 seconds for the computation. This provides an experimental basis of comparing the three approaches within the specified scenario. However, no general quantum supremacy is evident from this experiment, since the QAOA-based approach was executed using classical simulation and the traffic performance differences were noted from a single experimental setting.

The optimized decisions for traffic management were ultimately encoded into a JSON format that contains the diversions, signal timings changes, temporary bans, along with all of their parameters. This format will allow a uniform input channel for passing the output of the optimizer into the next step of traffic simulation. This allows the plan created by the optimizer to be tested separately from the baseline scenario.

This methodology thus includes event-demand modelling using scenarios, temporal arrival modelling, extraction of road networks from OpenStreetMap data, capacity modelling, binary decision modelling, QUBO modelling, Ising Hamiltonian modelling, optimization using QAOA by classical simulation, and traffic performance comparison. This framework offers a systematic means of examining how different route redirections, signal timing changes, and temporary traffic restrictions can be used in a predefined traffic scenario for a major event.

Quantum Traffic Optimizer - Traffic Decision and Simulation Architecture

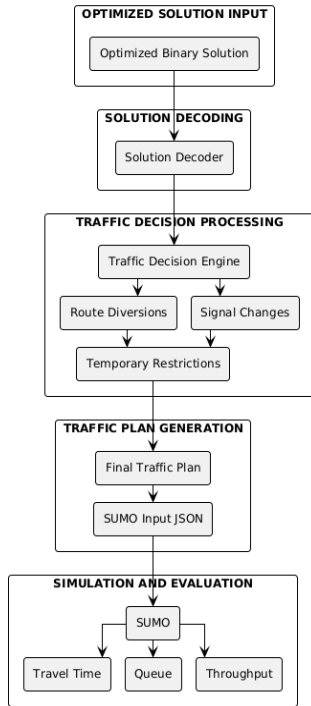


Figure 3: Traffic Decision and SUMO

5.2 Digital Twin

The system is a twin with a closed loop. It simulates traffic, people and trains of the particular district. No real-world data is included. All the data is generated by one simulator SUMO.

There are three layers:

1. Simulation layer (SUMO).

In this layer of road network, lights, cars, people and trains are represented. They all move in steps with step-time 0.5 sec.

2. Controller layer (Python: FastAPI, TraCI, NumPy, AsyncIO):

In this layer the Communication with SUMO via TraCI on a socket connection is done and management of the scenario with generation of crowd burst when train appears. Also calculation of the metrics and generation of JSON snapshot and sending the JSON snapshot via WebSocket with 30 FPS.

3. Visualization layer (Three.js, WebGL):

This layer works on the generation of city by requesting the snapshot and animation of objects .Moving of objects smoothly with LERP to provide 60 FPS and HUD display with KPIs and event phase.

The vehicles follow Intelligent Driver Model (IDM).The acceleration of the vehicle depends on the speed of the vehicle, the speed it wishes to attain, the distance to the vehicle in front, and the relative speed of the vehicle in front.

The sidewalks and plazas have stripes running along their lengths, each having a width of 0.55 m. Pedestrians prefer to walk on stripes overtake slow moving people avoid walking against the main flow of pedestrians and halt at signalized crossings.

The railway track is located at an elevation of 8.0 m on a viaduct supported by pillars after every 40 m. The trains follow the Krauss model and run on a schedule with fixed headways. Trains move slowly while approaching a station halt there for 30–35 s and accelerate again up to 80 km/h. The track is completely elevated. There are no level crossings. The road and rail collisions are avoided because of such construction.

SUMO uses X for East, Y for North, and Z for altitude. In SUMO, a heading of 0° indicates North direction. Increases clockwise. Three.js uses X for East, Y for Up, and Z for South, which is the direction where the viewer looks from. In Three.js, a heading of 0 radians indicates +Z direction. Increases counter-clockwise. The formulas are given as:

$$\text{Position: } (X_{3D}, Y_{3D}, Z_{3D}) = (X_{\text{SUMO}}, Z_{\text{SUMO}}, -Y_{\text{SUMO}})$$

-Eq(26)

$$\text{Rotation: } \theta_{3D} = -\theta_{\text{SUMO}} \pi / 180$$

-Eq(27)

Trains are displayed at an elevation of 8.1 meters. Cars and pedestrians are displayed at an elevation of 0.25 meters.

The Vehicle Motion: Intelligent Driver Model (IDM).
For vehicle a, when followed by a vehicle a-1:

$$dv/dt = a[1 - (v/v_0)^{\delta} - (s^*(v, \Delta v)/s)^2] \quad \text{-Eq(28)}$$

$$s^*(v, \Delta v) = s_0 + vT + v\Delta v/2\sqrt{ab} \quad \text{-Eq(29)}$$

$$s = x_{a-1} - x_a - L_{a-1} \quad \text{-Eq(30)}$$

$$\Delta v = v_a - v_{a-1} \quad \text{-Eq(31)}$$

Symbol	Meaning	Value
v_0	Desired speed	13.89m/s (Major roads) 11.11 m/s (Minor roads)
a	Maximum acceleration	2.6 m/s ²
b	Comfortable deceleration	4.5 m/s ²
Δ	Acceleration exponent	4
s_0	Distance at which jam occurs	2.0 m
T	Safe headway time	1.2 s

Table 2. Parameter Settings Used in Vehicle Motion.

Event Day demand is a piecewise non-homogeneous Poisson process that takes place in four phases.

Phase 1 Pre-event build-up: ($0 \leq t < T_1$):
 $\lambda(t) = L_{base} + (L_{surge} - L_{base}) * (t / T_1) \quad \text{-Eq(32)}$

Phase 2 Arrival surge: ($T_1 \leq t < T_2$):
 $\lambda(t) = L_{surge} + L_{stadium_inflow} \quad \text{-Eq(33)}$

Phase 3 Peak match: ($T_2 \leq t < T_3$):
 $\lambda(t) = L_{peak_match} \quad \text{-Eq(34)}$

Phase 4 Post-event egress: ($t \geq T_3$):
 $\lambda(t) = L_{base} + L_{egress} * \exp(-(t - T_3) / \tau) \quad \text{-Eq(35)}$

Constants:

$T_1 = 300$ s, $T_2 = 900$ s, $T_3 = 1500$ s. τ represents the depletion time constant.

The inflow to the Central Stadium Road and drop-off area is over 600 devices per hour.

The complexity of computations can be expressed as follows:

N_v - number of vehicles in motion. N_p - number of people on foot. E - number of street segments. J - the number of traffic lights. C - the number of clients connected. The update rate of the simulation is at 2 Hz (with an update interval of 0.5 s). The update rate of rendering is 60 Hz.

Space Complexity can be described by:

1. The usage of the memory on the server:

$O(N_v + N_p \cdot E \times H)$ (where $H = 120$ steps of history). The amount of memory occupied by SUMO = 45 MB. In combination with Python and metric memory usage the amount of memory is around 35 MB.

2. The amount of bandwidth consumed:

The size of the Event Day Snapshot Peak (250 vehicles and 300 people) = 18 KB. In the case of 30 Hz: 18 KB times 30 = 540 KB per second or 4.32 Mbps.

3. Memory consumption on the client's GPUs:

110 MB (considering vertex buffers, assembly buffers, and shaders.). Up-to-date methods of doing things come with some questions regarding their initial speed, scalability, and the application fields

themselves. Firstly, the problem lies in the TraCI communication overheads. Whenever different methods from the TraCI library are called - or `getAngle`, `getSpeed` and `getPosition` functions within TraCI are called, the number of exchanges N_v between the simulation application and the simulation program emerges. Thus, the overhead increases with the number of cars in the simulated system. However, the problem can be resolved through the `traci.vehicle.subscribe()` or `subscribeContext()` methods.

Another restriction that can appear is the overuse of JSON for communicating purposes. The usage of JSON suggests that certain CPU cycles will be used for serializing and parsing. What is more, the speed of the system is also decreased because of the number of actors.

Thus it is necessary to implement binary serialization systems such as Protocol Buffers or FlatBuffers because they allow cutting down on data and, accordingly, communication overhead.

Another problem with the -batched WebGL draw calls occurs within this particular system. The more cars and pedestrians are displayed through the meshes the more draw calls occur. It can reach up to 1500 draw calls per one frame. This is problematic for the end and mobile GPUs in terms of rendering performance. This can be solved by the implementation of the object pooling and GPU instancing technologies. For instance, the use of `THREE.InstancedMesh` allows rendering objects much better when their number is higher than 2000.

The other limitation is that it takes into account paths for pedestrians. The pedestrians have to stick to the paths provided to them and cannot change their location randomly within the large area around them. Consequently, pedestrian movement does not appear natural in the simulation. It can be addressed by applying either Social Force model or NavMesh technology.

Another flaw of the current model is the artificial generation of traffic demand. Since no calibration of any OD matrices takes place and there is not enough real-world ground truth statistics used in the simulation, it can be presumed that the generated traffic will not resemble real traffic at all. Thus, there is no way of validating the traffic statistics generated. However, the problem can be solved by the use of open and calibrated OD data along with validation of simulated traffic statistics via sensors.

Finally, the model is run on the basis of a feeder assumption. The traffic enters the simulation network through certain edges like , , etc. Traffic, that is outside the simulated region, does not appear directly in the simulation; thus, all external traffic interactions and traffic spillbacks are not completely simulated. This issue in the model could be resolved either by expanding the boundaries of the network or by connecting the simulation to a regional traffic simulation model.

6. RESULT

The traffic optimization framework proposed for the event was assessed based on five different scenarios characterized as good weather, heavy raining, movement of VIPs, road construction, and combinations of various situations. The framework was assessed by using three optimization techniques, namely greedy optimization, classical optimization, and QAOA-based optimization.

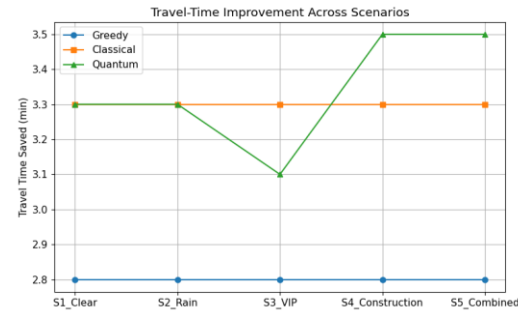


Figure 4: Travel Time Improvement Access Scenarios

The greedy was constant at 2.8 minutes saved for every experiment. Classical was constant at 3.3 minutes for all experiments. QAOA was 3.3, 3.1, 3.5 minutes for the clear/heavy-rain, VIP, construction/combined cases respectively. QAOA was equivalent to classical for clear/heavy-rain, 0.2 minutes worse for VIP, and 0.2 minutes better for construction/combined scenarios.

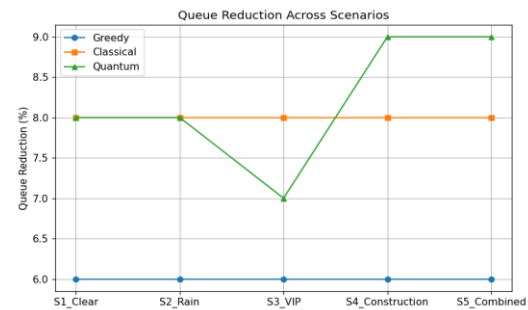


Figure 5 : Queue Reduction Across Scenarios

The second metric considered was reduction of queues. When using the greedy approach, the average number of queues reduced per case was 6%. The classical approach averaged 8% reduction overall. The QAOA based approach achieved 8%, 7%, and 9% reduction under clear and heavy-rain, VIP, and construction and all scenarios, respectively.

When used with the case based approach to manage queues, the QAOA approach achieved 8% reduction compared to the classical approach when managing queues in all conditions and scenarios. QAOA achieved 9% reduction with a case based approach in the combined scenario whereas the classical approach achieved only 8%. It appears as though the two approaches are able to handle the three operational constraints in the same way. The travel-time results for scenario-level cases lend themselves to the same conclusion as the queue results.

When the classical approach is used the travel-time is 3.3 min in every case. When the QAOA based approach is used in every case it is 3.1 min in the VIP case and 3.5 min in every other case. There is a difference in the case based approach to manage queues for the QAOA approach to manage queues. Overall the three approaches used in this work each gained 2.80, 3.30, and 3.34 min in average travel-time savings in all scenarios and gained 6.0, 8.0, and 8.2% in queue reduction for greedy, classical, and QAOA-based approaches, respectively.

Compared to the greedy baseline, QAOA achieved average improvements of 19.286% in travel-time saving and 36.667% in queue reduction, whereas the classical approach achieved 17.857% and 33.333%, respectively. All means are directly reported in the efficiency report. Improvements of QAOA over the classical approach were small. 0.04 min in average travel-time saving and 0.2 percentage points in average queue reduction. So the main conclusion is that the QAOA-based approach reached traffic-performance results very close to the classical approach, rather than algorithmic dominance.

The QUBO objective values were also investigated to assess the behavior of the mathematical optimization formulation. The greedy and classical objective values were constant at five scenarios at approximately -528.28 and -601.68, respectively. On the other hand, the QAOA-based objective changed with the scenario, taking values of -469.25 for the clear scenario, -489.00 for heavy rain, -461.20 for the VIP scenario, -489.50 for construction, and -425.25 for the combined scenario.

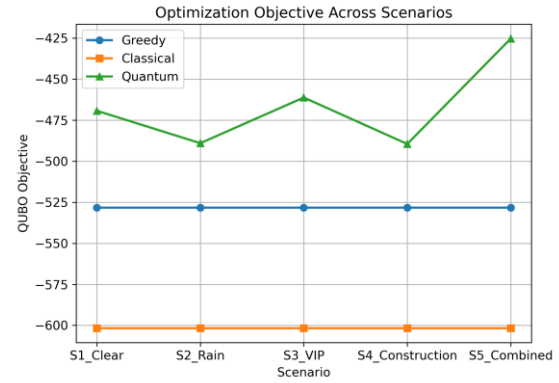


Figure 6 : Optimization Objective Across Scenarios

The average QUBO objective values were -528.28 for greedy, -601.68 for classical and -466.84 for the QAOA-based approach. Since the QUBO objective represents the objective of the mathematical optimization problem, these values have to be considered separately to traffic-performance indicators, as travel-time saving and queue reduction. In the current formulation, the classical method achieved a numerically lower QUBO objective, while the QAOA-based method achieved competitive traffic-performance outcomes. This shows that the optimization-objective value and downstream traffic-performance indicators are related but non-interchangeable evaluation metrics.

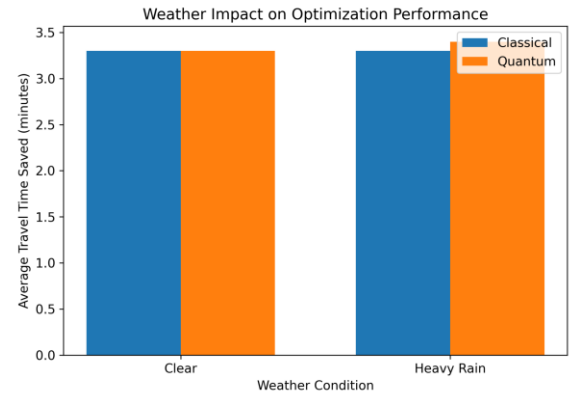


Figure 7 : Weather Impact on Optimization Performance

The impact of weather conditions was analyzed by comparing the clear and heavy-rain scenarios. For the clear setting, the classical and QAOA-based approaches both resulted in 3.3 minutes of travel-time saving. The same was observed for the heavy rain condition, where the classical approach produced 3.3 minutes of travel-time saving, and the QAOA-based

method produced 3.3 minutes as well. Correspondingly, both the approaches reduced the queues by 8% each in both the settings.

Based on the weather-specific analysis, it can be concluded that the QAOA-based approach was able to maintain the same level of traffic-performance as that of the classical approach when the scenario was switched from clear weather to heavy rain. This provides evidence that the optimization formulation was able to remain stable under the environmental change that was encapsulated within the experimental scenarios.

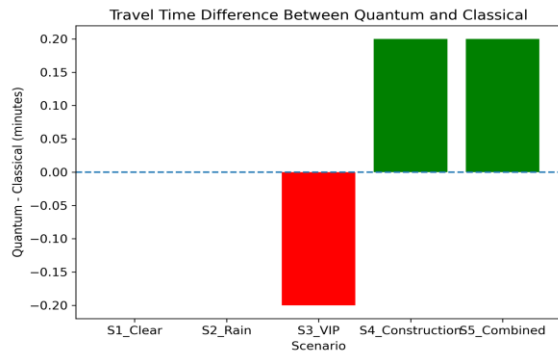


Figure 8: Travel Time Difference Between Quantum and Classical Algorithm

A direct comparison between the QAOA-based and the classical optimization further highlighted these scenario-dependent results. The differences in travel time was zero for the clear and heavy-rain scenarios, 0.2 min for the VIP scenario, and +0.2 min for the construction and combined scenarios. Therefore, the QAOA-based optimization was neither better nor worse than the classical optimization but instead depended on the constraints being applied in the scenarios.

This was especially true in the combined scenario where the maximum number of simultaneous constraints was present, and in which the QAOA based gave a travel-time saving of 3.5 min and queue reduction of 9% compared with 3.3 min and 8% for classical optimization, corresponding to QAOA based gave the highest values of trafficperformance for QAOA in this study.

The experimentally observed outcomes suggest that the QAOA-based formulation successfully obtained performance equivalent to classical optimization across the five pre-event traffic scenarios. The QAOA-based optimization resulted in an average of 3.34 min saving in travel time (compared to 3.30 min for the

classical optimization) and an average of 8.2% reduction in queue length (compared with 8.0%). Based on the classical optimization method, QAOA achieved a 19.286% improvement in terms of the average travel-time saving, and a 36.667% improvement in terms of the average queue reduction over the greedy method. At the level of individual scenarios, the QAOA based achieved the same outputs as classical method in the clear and heavy-rain scenarios, led to slightly lower traffic management outputs in the VIP scenario, but outperformed classical optimization in the combination of the two scenarios (QAOA gave 3.5 min travel-time saving and 9% queue reduction whereas classical optimization gave 3.3 min and 8%, respectively).

Analysis of the computer simulation demonstrated a different trade-off than in the scenario analysis. The QAOA implementation took an average of 18.03 seconds, versus 2.55 seconds for the classical optimization, and had a lower overall efficiency score than the classical implementation. That result makes sense given that the current QAOA implementation was tested on a classical computer simulation rather than a real quantum computer.

The Digital Twin was built for the MA Chidambaram Stadium (Chepauk) region using SUMO as the microscopic traffic engine. The network covers the main roads, junctions, traffic signals, pedestrian paths, and the public-transport elements around the stadium. Each vehicle in SUMO moves on its own, so queues and congestion form from individual driving behavior and not from fixed averages. Figure 9 shows the SUMO network used in the experiments. The stadium is marked in the middle of the study area, with the major roads forming a loop around it. This layout makes the region a good test case, because event traffic has to pass through a small number of junctions.

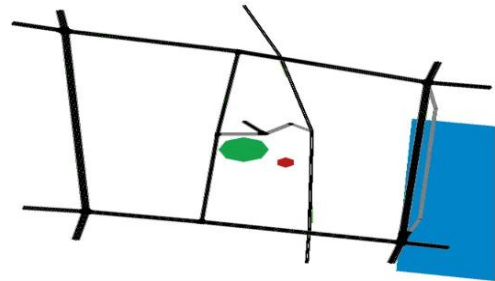


Figure 9: SUMO-based Chepauk Digital Twin network.

Two scenarios were prepared, a Normal Day and an Event Day. The Event Day scenario represents the

heavier traffic seen when a match is on. The demand can be set from the interface for each simulation hour. For the Event Day run, the inputs were 5,540 cars, 45 buses, 1,550 two-wheelers, 2,400 pedestrians, and 80 local trains. These values scale the existing SUMO flow definitions, while the route geometry stays the same. This keeps every run comparable. All results in this section come from the Event Day scenario.

The image shows a 'SCENARIO INPUT' panel with a 'Configure traffic' tab. Under the 'Scenario' section, 'Event day' is selected. Below, 'Demand per simulation hour' is shown with input fields for different vehicle types: Cars (5540), Buses (45), Two wheelers (1550), Pedestrians (2400), and Local trains (80). A note at the bottom states: 'Values scale the existing SUMO flow definitions; route geometry stays unchanged.'

Figure 10: Scenario input panel showing Event Day demand per simulation hour.

SUMO produces the traffic state at every simulation step, and TraCI reads this state and passes it to a Python backend. The state includes vehicle positions, traffic-signal conditions, pedestrian activity, and congestion information. The backend then streams it over WebSocket to a Three.js interface, which redraws the 3D scene each time a new frame arrives. During testing, the WebSocket delivered both the road geometry and live simulation frames. The geometry is sent first so the browser can build the road network, and the live frames follow to update the moving elements. One recorded frame, taken at about 437.5 seconds of simulation time, held 260 vehicles, 141 pedestrians, 7 traffic lights, and 60 congestion edges. This confirms that the interface is driven by the running simulation and not by a prepared animation. The numbers also change from frame to frame as traffic builds up.

Figure 11 shows the complete dashboard. The classical baseline and the optimized simulation run side by side, with synchronized cameras. The cameras move together, so the viewer looks at the same location in both views and can see where the two states differ. At the clock reading of 04:26.0, the network held 236 vehicles and 86 pedestrians. The vehicle mix was 174 cars, 3 buses, 57 two-wheelers, and 2 local trains. The average speed was 13.2 km/h, the queue length was 617.5 m, and 17 roads were congested. Road colors follow four congestion levels, from low to

severe, so crowded sections are easy to tell apart from free-flowing ones. The congestion data comes from SUMO itself and is not drawn by hand. The side panel also shows the live metrics, and the bottom panel shows the optimization status and the actions applied.

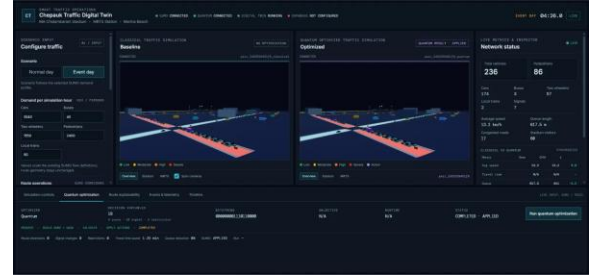


Figure 11: Digital Twin dashboard with the baseline and quantum-optimized simulations, live metrics, and optimization status

The Digital Twin does more than display traffic. It can also act on the running simulation. The optimization problem used 18 decision variables: 4 for routes, 10 for signals, and 4 for restrictions. The solver returned the bitstring 000000001110110000, and the dashboard reported the run as completed and applied.

In the live integration experiment, 23 traffic-control commands were generated, and every target identifier matched an element in the active SUMO network. After the commands were applied, route-related values changed on 18 mapped edges. Signal timing also changed. The next-switch time moved from 36 to 42 seconds at J_CENTRAL_STAD, from 36 to 46 seconds at J_NE, and from 36 to 43 seconds at J_SW.

The optimization model uses logical labels such as Anna Salai and J1, which SUMO does not understand. A mapping layer converts these labels into real SUMO identifiers before any command is issued. All four optimization corridors and the J1–J10 signal identifiers were checked against the Event Day network. This keeps the optimization model separate from the simulation itself.

Figure 12 shows the full sequence. The baseline traffic is observed, optimization is triggered, and the result is turned into traffic actions. These actions become TraCI commands, SUMO updates its state, and the new state is shown in the interface.

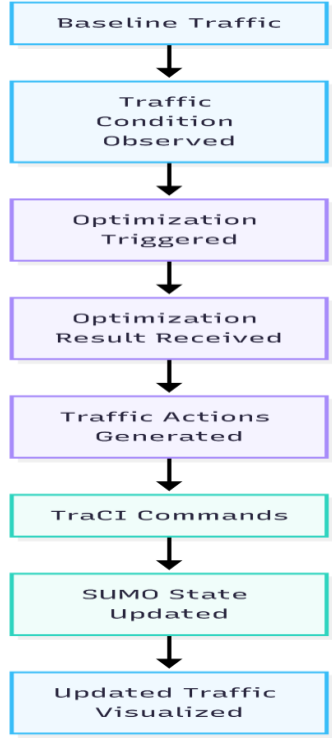


Figure 12: Operational sequence from baseline traffic to the updated visualization.

The side-by-side comparison in Figure 11 shows the effect of the applied result. The average speed stayed at 13.2 km/h in both views, while the queue length fell from 617.5 m to 611 m, a drop of 6.5 m. The dashboard also reported a travel-time saving of 1.25 minutes. These gains are small in this snapshot, which was taken early in the Event Day run. At that point, the network had not yet reached its heaviest load. The main finding here is that the full loop works. A decision made by the optimizer reaches the simulation and produces visible changes. Larger gains can be studied in longer runs and at peak event demand.

The Unicorn-based server started successfully. Its endpoints for geometry, optimization triggering, optimization-run information, and the latest optimization plan all returned successful responses. This shows that the backend is ready to be used by the interface and by the optimization workflow. The browser opened the WebSocket connection, rendered the 3D road network, and received the quantum optimization information through the same state stream. Because everything travels over one stream, the interface always shows the same state as the simulation.

The results confirm three working stages. SUMO generates the traffic behavior, TraCI reads and changes the running simulation, and WebSocket with Three.js turns that state into a live visual interface. Commands from the optimization workflow reached SUMO and changed both route values and signal timings. The identifier mapping also worked on the Event Day network, so abstract optimization labels were applied to real SUMO elements. This makes the Digital Twin a usable platform for testing traffic-management decisions around the Chepauk stadium during events. It gives a safe place to try a decision and see its effect before it is used on real roads.

7. DISCUSSION

7.1 Overall Findings

The framework was tested on an IPL event around the M. A. Chidambaram Stadium, analyzing route diversions, signal extensions and temporary traffic restrictions. QAOA yielded a feasible traffic-management plan close to classical optimization.

7.2 Strengths of the Proposed Framework

The framework brings different traffic-management decisions into one optimization problem. Route diversions, signal extensions, and temporary restrictions can therefore be considered together. The QUBO also includes operational constraints through penalty terms, which helps avoid invalid combinations of decisions. The selected decisions can then be exported for further traffic-simulation evaluation.

7.3 Limitations

In this work, we analyze the pre-event traffic planning based on a target event scenario and several fixed traffic-management measures such as route diversion, signal extension, and road restriction. The assessment was carried out in a simulated quantum environment using synthetic demand, which makes it possible to conduct the effectiveness of our proposed framework. In the future, we can expand the scope by simulating multiple event scenarios, including larger and more varied road networks, live-traffic conditions, and physical quantum hardware.

7.4 Practical Implications

The framework can be used for pre-event traffic planning by allowing different combinations of route diversions, signal adjustments, and temporary restrictions to be evaluated before an event. The selected decisions can also be exported for further testing through traffic simulation. This can help compare alternative traffic-management plans before they are considered for real-world use .

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